8 November 2026

Dear Editor,

We thank the Editor and the three Reviewers for their careful reading of our manuscript, "Evidence of fault zone transport of road-salt contaminated groundwater from highways to streams" (Manuscript ID: 101225), and for their constructive comments. Below we reproduce each comment in *gray italics*, followed by our response and the corresponding change to the manuscript in blue. A marked-up version showing all changes relative to the original submission accompanies the clean revised manuscript. We made the marked-up version with latexdiff, which does not colour the contents of tables read in from separate files. Table 1, Table 2 and the Appendix A table are new or fully replaced, so their contents appear without highlighting; their captions are marked. Author names and order are unchanged.

The reviewers' central, shared concern was that the strength of our claims exceeded what the data support. We agree, and the revision is built around that point. The principal changes are:

- The manuscript is reframed as a *hypothesis-generating case study*, and the title now poses the question the study addresses: "Do fault zones route road-salt-contaminated groundwater to streams? A Hudson Valley, NY Case Study". With the corrected and expanded analyses described below, our answer from this survey is that the evidence is inconclusive: one fault-zone contrast shows a small, significant increase in specific conductance, but it is gradual rather than abrupt, and no change point coincides with a fault contrast boundary. The revised paper reports this plainly, sets out what a low-cost longitudinal survey can and cannot resolve, and proposes a specific design for testing the fault-conduit hypothesis (Section 4.5).
- In re-examining the statistics as the reviewers asked, we found that three of the four Site 3 zones combined measurements from two survey days (19 and 26 October 2018) whose conductivity differed by 23.5–29.7 μS cm^{-1}. We now compare zones only within a single survey day. This substantially reduces the Site 3 fault-zone step (Contrast E), which we report transparently. The same re-examination showed that the 2020 chloride peaks lie near a different mapped fault (Contrast D) rather than at Contrast E, and we have corrected that attribution.
- We added change-point / segmented-regression analysis of specific conductance against distance (new Figure 6), an autocorrelation-adjusted assessment of the zone contrasts, an expanded Table 1 (means ± SD, confidence intervals, degrees of freedom, Mann–Whitney results), a new Table 2 (adjusted p-values and distance to the nearest major road), the same change-point analysis applied to the chloride transect, and an appendix (Appendix A) listing every measurement.
- The z-score normalization has been removed; all analyses use specific conductance in μS cm^{-1}.
- The Introduction is restructured to reach fault hydrogeology earlier, with the conduit–barrier literature for fractured crystalline and metamorphic rock (including Caine et al., 1996, and Bense et al., 2013), a paragraph on tributary mixing, and references for each testable prediction.
- A new Limitations section (Section 4.4) and an expanded discussion of alternative salt sources (Section 4.2: road proximity, developed land, wastewater plants, diffuse groundwater). Permitted wastewater facilities are now shown on the maps (Figures 1 and 3).
- All six figures are revised or new (contrast labels, flow direction, the Z–Z′ transect, survey-date symbology and an inset, a simplified overview map, and a new change-point figure). We found no citable positional accuracy for the mapped faults, so their uncertainty is discussed in the text rather than drawn as an error zone.

- The management recommendation is replaced by a statement that mapped faults are at most a candidate screening layer to be tested (Section 4.6), and references now carry DOIs where they exist, with grouped citations sorted and compressed.

We were not able to collect additional field data for this revision. Where a reviewer's suggestion would require new measurements (discharge, geochemical tracers, repeated cross-channel sampling), we state this explicitly as a limitation and as a priority for future work.

Sincerely,

Tadd Bindas and Zion Klos

Response to the Editor

Comment E.1

> *Overall, I consider the manuscript relevant and potentially suitable for publication, and therefore recommend publication provided that the points raised above are adequately addressed.*

Response: We thank the Editor for this assessment. We have addressed every point raised by the reviewers, as detailed below, and provide a marked-up manuscript highlighting all changes. Author names and order are unchanged; the affiliation of the first author is now given as Marist University, with his present address noted in a footnote.

Response to Reviewer 1

Comment R1 (general)

> *The topic is novel and opens a useful and interesting discussion to associate structural geology directly to the freshwater salinization problem. The writing is generally clear and good. My main concern is that with the current structure, the data and statistics presented do not support the strength of the claims. The SC steps coincide with mapped faults at two crossings. However, the paper still does not show that the faults cause these steps, that the faults connect to a highway, or that road salt rather than another source is responsible. That's why I'd recommend to reframe the work as a hypothesis-generating case study, discuss alternative sources and the study's limitations explicitly in the text, clarify the maps and temper the conclusions.*

Response: We thank the Reviewer for this clear summary and agree with it. Our data show where specific conductance changes along the channel; they cannot establish that faults cause those changes, that the faults are hydraulically connected to the highways, or that road salt rather than another source is responsible. We have followed each of the Reviewer's recommendations: the manuscript is now framed as a hypothesis-generating case study, and the title has been changed from a statement of evidence to the question the study poses; alternative sources are discussed in an expanded section (Section 4.2); a new Limitations section (Section 4.4) lists the gaps explicitly; all maps are revised; and the conclusions are tempered throughout.

Changes in manuscript:

> Title changed to "Do fault zones route road-salt-contaminated groundwater to streams? A Hudson Valley, NY Case Study"; Abstract, Introduction (final paragraph), Discussion and Conclusions rewritten in hypothesis-generating language. Section 1: "The study is a hypothesis-generating screening case study: it can show whether the predicted pattern is visible in a low-cost survey, and what a conclusive test would require, but it cannot by itself demonstrate that faults transport road salt." New Section 4.4 "Limitations"; expanded Section 4.2; Figures 1–5 revised and Figure 6 added.

Comment R1.1

> *The Introduction reaches geology, the paper's central argument, only after long sections on impacts and the global salinization syndrome*

Response: Agreed. We have restructured the Introduction so that fault hydrogeology follows directly after a condensed account of road salt and its groundwater legacy. The material on ecological impacts has been shortened, and the freshwater salinization syndrome is now introduced in two sentences after the fault paragraph; its implications are discussed in Section 4.3.

Changes in manuscript:

> Section 1 reorganized: (1) road salt and its impacts (condensed); (2) the groundwater legacy; (3) faults as conduits and barriers in fractured bedrock (moved forward, expanded); (4) salinization syndrome (two sentences); (5) tributary mixing (new); (6) hypothesis and testable predictions.

Comment R1.2

> *In the Introduction soil (not relevant here when looking specifically for chloride) and groundwater timescales are mixed together*

Response: Agreed. We have removed the soil-persistence statement and now describe only the groundwater storage and release timescales relevant to baseflow chloride.

Changes in manuscript:

> Section 1: sentence citing soil persistence of 2.5–5 months removed; the groundwater storage and release argument ("released to streams over years to decades") is retained.

Comment R1.3

> *The three testable predictions have no supporting references*

Response: We have added supporting references for each prediction: that point groundwater inputs produce localized steps in longitudinal specific conductance; that late-season baseflow is dominated by groundwater; and that chloride is the most direct tracer of deicing salt.

Changes in manuscript:

> Section 1, final paragraphs: citations added for each prediction (Harte & Trowbridge, 2010; Hodgins et al., 2024, 2026; Miller et al., 2014; Mullaney et al., 2009).

Comment R1.4

> *The fault hydrogeology rests on a karst analogue and a study of unconsolidated aquifers, not on work in metamorphic bedrock (e.g., Caine et al., 1996; Bense et al., 2013).*

Response: We thank the Reviewer for these references. The fault-hydrogeology framing now rests on the fault-zone architecture model of Caine et al. (1996) and the review of Bense et al. (2013), together with studies of fault-controlled flow in fractured crystalline and metamorphic bedrock (Manda et al., 2008; Kim et al., 2014). The karst example is retained only as an explicitly labelled illustration of how anisotropic fault-parallel flow can be, and we note that carbonate aquifers are not a direct analogue for the metasedimentary bedrock of our study area.

Changes in manuscript:

Sections 1, 2.1 and 4.3: Caine et al. (1996), Bense et al. (2013) and two additional bedrock references (Manda et al., 2008; Kim et al., 2014) added. Section 1: the karst example is "cite[d] only as a carbonate example of how anisotropic fault-parallel flow can be, because karst is not a direct analogue for the metasedimentary bedrock of our study area."

Comment R1.5(1)

A longitudinal survey of specific conductance as in the paper is a reasonable, low-cost way to screen for groundwater inputs. It can show where SC changes, but it is difficult to attribute those changes specifically to faults or to road salt so I'd recommend being careful in state that, more specifically: (1) There are no discharge data and no source tracers

Response: We agree, and we now describe the survey as a screening tool that shows where specific conductance changes rather than attributing the changes. The absence of discharge data and source tracers is stated explicitly in the new Limitations section (Section 4.4); without them we cannot estimate the volume of any input or attribute salt to roads, wastewater or other sources. Paired discharge measurements, chloride-to-bromide ratios and groundwater age tracers are identified as the next step for testing the hypothesis (Section 4.5).

Changes in manuscript:

Sections 1 and 4.1 (screening framing); Section 4.4 (Limitations): "We have no discharge measurements ... We have no source tracers, so we cannot attribute salt to roads, wastewater or other sources." Section 4.5 (test design).

Comment R1.5(2)

(2) There is only one chloride transect, measured with an ion-selective probe

Response: Agreed. We now state that the chloride data are a single transect of 18 main-stem samples (plus one side-tributary sample), measured with a YSI Pro Series 1005 chloride ion-selective electrode. We give the manufacturer's stated accuracy (±15% of reading or $\pm 5\,\mathrm{mg\,L^{-1}}$, whichever is greater), which at our concentrations (about 14–50 $\mathrm{mg\,L^{-1}}$) is ±5 to $\pm 7.5\,\mathrm{mg\,L^{-1}}$. Only the larger differences along the transect exceed it, and we list both the single transect and this measurement uncertainty among the limitations. The transect, collected in a different year from the conductivity survey, is presented as a separate, descriptive observation rather than as confirmation (see also R2.7).

Changes in manuscript:

Section 2.2: "Chloride was measured in the stream with a YSI Pro Series 1005 chloride ion-selective electrode (YSI 605105) ... an accuracy of ±15% of reading or ±5 mg/L, whichever is greater"; Section 3.2 (the transect is "a single transect from a different year ... so it cannot confirm the conductivity pattern"); Section 4.4 (single transect and measurement uncertainty).

Comment R1.5(3)

(3) Comparing zone means with t-tests cannot show whether a change is abrupt. Change-point analysis or segmented regression of SC against distance would test this directly and could be recommended

Response: We thank the Reviewer for this suggestion and have implemented it. For each single-day survey transect we fitted continuous piecewise-linear regressions of specific conductance against downstream distance, with zero, one or two breakpoints chosen by BIC, and checked them with a penalized change-in-mean analysis (PELT). No breakpoint in specific conductance coincides with a fault contrast boundary or with a point where the digitized trace crosses the transect. At Site 2 a single breakpoint is selected about 1.6 km upstream of the Contrast B boundary, after which conductance increases steadily downstream (from 0.1 to 1.6 μS cm^{-1} per km), so the increase across Contrast B is better described as a gradual rise than an abrupt step. No breakpoint is supported for Site 1 or for the 19 October survey at Site 3, and on the upper 26 October Site 3 transect two breakpoints are only marginally preferred and lie away from any trace. On the lower 26 October Site 3 survey the only breakpoint lies about 520 m upstream of the Contrast E zone boundary, 123 m from the nearest mapped trace, whose position is itself uncertain (see R2.6). We report these results as showing that, in this survey, changes in specific conductance are not concentrated at the mapped faults, and we have revised the conclusions accordingly.

Changes in manuscript:

> Section 2.3 (new method paragraph); Section 3.3 (new results): "Segmented regression and a change-in-mean check on each single-day transect ... find no specific-conductance breakpoint at a fault contrast boundary or at a point where the digitized trace crosses the transect." New Figure 6 shows specific conductance against distance for each single-day transect, with fitted segments, breakpoints, and the positions of trace crossings and tributaries.

Comment R1.6

> *Table 1 lacks standard deviations, degrees of freedom, confidence intervals, and the Mann-Whitney results. I'd add that to help the reader interpret the results*

Response: Agreed. Table 1 now reports, for each contrast, the survey date, zone sample sizes, zone means ± standard deviation, the difference in means with its 95% confidence interval, the Welch t statistic, Welch–Satterthwaite degrees of freedom, p-value, and the Mann–Whitney U statistic and p-value. The autocorrelation-adjusted p-values are in the new Table 2 (see R1.8).

Changes in manuscript:

> Table 1 regenerated from the source data (`src/within_day.py`) with the columns above; Table 2 added.

Comment R1.7

> *The reported t values imply a Zone 3-2 standard deviation larger than the contrasts being tested*

Response: The Reviewer is correct, and this observation led us to an error in our zone definitions. Zones 3-1, 3-2 and 3-3 each combined measurements from two survey days (19 and 26 October 2018), between which the overall conductivity of the stream differed by 23.5–29.7 μS cm^{-1} (eleven pairs of points within 120 m; two measurements 18 m apart at the seam between the surveys read 332.0 and 355.5 μS cm^{-1}). Within a single day the Site 3 zone standard deviations are 0.4–2.6 μS cm^{-1}; pooling the two days inflated them to 12.6–15.8 μS cm^{-1}. The original statement that every contrast was computed within a single day's data was incorrect, and we apologize for it. We now compare zones only within a single survey day. With this correction, the Site 3 fault-zone contrast (Contrast E) falls

from +18.0 to +1.6 μS cm^{-1} (95% CI −0.3 to +3.4; Welch $p = 0.090$) and is no longer significant. We report this in full, and the Discussion now uses it to illustrate how a survey spread over several days without repeated tie points can produce a spurious step.

Changes in manuscript:

Sections 2.2, 2.3 and 3.1: zone contrasts recomputed within survey days; the incorrect statement removed. Section 3.1 ("Pooling days"): "When the two survey days are pooled, Contrast E appears as a rise of +18.0 μS/cm ($p < 0.001$), larger than any other step in the survey. That value is an artefact of pooling two survey days." Table 1 and Figure 4 updated (survey date shown); Sections 4.5 and 5 use the case as a pitfall to avoid.

Comment R1.8

Spatial autocorrelation between samples 100 m apart is acknowledged but not handled, so the p-values are likely too small

Response: Agreed. We now estimate the lag-1 autocorrelation of specific conductance along each transect and use it to compute an effective sample size for each zone, and report the t-tests with these adjusted sample sizes. The adjustment raises the p-value wherever the residuals are positively autocorrelated (Contrasts B, D, E, and C on 26 October) and leaves the others unchanged; Contrast B remains significant ($p = 0.017$ with adjusted sample sizes, compared with $p < 0.001$ unadjusted), and Contrasts D and E are further from significance ($p = 0.28$ and 0.24). We also note that, because neighbouring samples are not independent, the p-values should be read as approximate, and that the change-point analysis (Comment R1.5(3)) gives a complementary test that does not treat the samples as independent replicates.

Changes in manuscript:

Section 2.3 (autocorrelation method; $n_{\mathrm{eff}} = n(1-\rho)/(1+\rho)$): "we treat the adjusted p-values as a sensitivity check rather than a correction." Table 2 (adjusted n_{eff} and p); Section 3.1.

Comment R1.9

The z-score normalization that was presented on the Methodology has no effect on within-site tests

Response: The Reviewer is correct. All tests were carried out on specific conductance in μS cm^{-1}, and the normalization described in the Methods was a remnant of an earlier version of the analysis. We have removed the normalization and its equation; the figures now also show specific conductance in μS cm^{-1}.

Changes in manuscript:

Section 2.3: the normalization equation and paragraph removed (the remaining equation is the Welch t statistic); Figures 2–4 colour scales changed to μS cm^{-1}.

Comment R1.10

The explanation for the non-significant Contrast D is incorrect if you consider that two other contrasts have the same sample-size split

Response: The Reviewer is right: in the submitted analysis Contrasts C and E had the same 17/6 split, so unequal sample sizes cannot explain the result at Contrast D. We have removed that explanation. Within a single survey day Contrast D shows a small, non-significant decrease ($-2.8\,\mu\mathrm{S\,cm^{-1}}$, 95% CI -7.4 to $+1.7$, $p = 0.14$), and we report it as a fault with no tributary that shows no increase in specific conductance. We also note that the 2020 chloride peak lies near the survey points flagged for Contrast D (133–468 m from the nearest flagged point; see R2.7), but it rests on one sample from a different year, and we describe it as an observation worth testing, not as support for the hypothesis.

Changes in manuscript:

Sections 3.1, 4.1 and 5: sample-size explanation removed; Contrast D reported as showing no increase. Section 4.1: "Contrast D, at a fault with no tributary, shows no increase." Section 5: "A second fault shows no increase."

Comment R1.11

Some steps, such as 4.3 µS/cm at Contrast B, are close to the uncertainty of temperature-compensated SC

Response: Agreed. We now state the instrument accuracy. The manufacturer gives no separate accuracy for the Orion 013010MD cell; its cell constant is set by calibration. The compatible Orion Star meters specify ±0.5% of reading ±1 digit, i.e. about $\pm 1.7\,\mu\mathrm{S\,cm^{-1}}$ at $310\,\mu\mathrm{S\,cm^{-1}}$ (Site 2) and $\pm 1.9\,\mu\mathrm{S\,cm^{-1}}$ at $360\,\mu\mathrm{S\,cm^{-1}}$ (Site 3). All Site 2 measurements were taken on one day with the same instrument and a narrow water-temperature range (15.4–15.8 °C), so a calibration offset would affect both zones equally; the within-zone standard deviations (1.1 and $1.9\,\mu\mathrm{S\,cm^{-1}}$) are smaller than the $4.3\,\mu\mathrm{S\,cm^{-1}}$ difference. The $4.3\,\mu\mathrm{S\,cm^{-1}}$ difference at Contrast B is about 2.5 times this accuracy, whereas the $1.6\,\mu\mathrm{S\,cm^{-1}}$ difference at Contrast E is within it. We nonetheless describe Contrast B as a small change near the limit of what the survey can resolve, and we have tempered its interpretation.

Changes in manuscript:

Section 2.2 (instrument accuracy); Sections 3.1, 4.1 and 4.4 (Contrast B described as "a real but small difference"; Contrast E "within the instrument's stated accuracy").

Comment R1.12

The measured SC differences appear genuine and are within an expected range, but the interpretation goes beyond them. Other sources could fit the pattern at least as well as the faults so I'd be careful in how to state that specially considering that the highest-SC zones sit just beside US 44 / NY 82 (from the map), Site 2 runs beside developed land, A sewage treatment plant and septic systems lie upstream (according to the same study site evaluated in Kelly et al., 2008) and groundwater unrelated to faults can enter the creek anywhere.

Response: We agree, and Section 4.2 now discusses each of these alternatives: road proximity, including salt stored in roadside soils and shallow groundwater; developed land (Site 2 runs beside US 44 and ends near the settled area around Titusville); wastewater; and diffuse groundwater discharge unrelated to faults. On wastewater, we located the permitted facilities from EPA ECHO records. The Millbrook sewage treatment plant described by Kelly et al. (2008), whose study for the East Branch of Wappinger Creek attributed about 4% of the sodium chloride input to sewage

(septic systems and the plant together), is upstream of all three sites: about 2.4 km along the channel above Site 3, and further above Sites 1 and 2. As a continuous upstream source it would raise baseline conductance but would not by itself create a step at a zone boundary. More relevant to Contrast B, two small private plants (Stratford Farms and Ennis Mobile Home Park) have listed locations 240–370 m from Zone 2-1, just upstream of the Site 2 breakpoint. We could not establish their outfall locations or flows, so we cannot test whether they contribute, and Section 4.2 now names them as a plausible, untested alternative source for the Site 2 increase. The Titusville plant lies just downstream of the last Site 2 survey point, so it cannot affect Contrast B, and the Valley Dale plant discharges to a tributary about 2 km away. The facilities are shown on Figures 1 and 3. We now state that our data cannot distinguish these sources from fault-routed groundwater, and that resolving them would require tracer and discharge measurements. We have also quantified the distance from each sampling point to the nearest major road using existing road data (TIGER/Line), and report the median for each zone descriptively (245–927 m; Table 2) without attributing the contrasts to it. The Site 3 zones lie 245–425 m from the nearest major road, mainly US 44, with no monotone relation to the conductance pattern. At Site 2, for example, the downstream zone is farther from a major road than the upstream zone (median 726 vs. 335 m), so road proximity does not by itself account for the increase at Contrast B, although it cannot be excluded either.

Changes in manuscript:

> Section 4.2 expanded (road proximity, developed land, wastewater, diffuse groundwater); Table 2 (road distances); Figures 1 and 3 (wastewater facilities). The "conserved forest" argument for discounting land-use inputs is withdrawn: "we withdraw it."

Comment R1.13

> *Several statements overreach, such as "strong initial evidence" and "this newly observed process". Section 4.4 says all three crossings point to a link with fault zone transport, although Contrast D shows a non-significant decrease. Recommending different deicers (in the Conclusions) wherever a road crosses a mapped fault is premature on three crossings in one lithology. The Discussion also offers little comparison with other longitudinal salinity surveys or with fault hydrogeology in metamorphic rocks.*

Response: Agreed on all points. We have removed or rephrased the overreaching statements, including those quoted; the Discussion and Conclusions no longer claim that all three crossings support the hypothesis (the Reviewer's Section 4.4, "Suggestions for future research" in the original submission, no longer makes that claim; the discussion of the three contrasts is now in Section 4.1, where Contrast D "shows no increase"); and the management section (Section 4.6) now proposes mapped faults as a candidate screening layer to be tested, rather than recommending changes in deicing practice. We have added comparisons with other longitudinal salinity surveys and with fault hydrogeology in crystalline and metamorphic rocks (e.g. Harte & Trowbridge, 2010; Hodgins et al., 2024, 2026; Orr & Boyer, 2026; Manda et al., 2008; Kim et al., 2014).

Changes in manuscript:

> Sections 4.1, 4.6 and 5 revised; new comparison paragraphs in Section 4.3. Section 4.6: "The evidence here does not justify changing deicing practice on the basis of fault locations."

Comment R1.14

some references lack DOIs or report numbers and grouped citations are out of order.

Response: Corrected. DOIs have been added to the references where they exist. Four entries (Budnik et al., 2010; Kelly et al., 2019; Kelting et al., 2010; Adirondack Road Salt Reduction Task Force, 2023) are reports without DOIs; they carry report numbers or URLs. Grouped citations are now sorted and compressed.

Changes in manuscript:

Reference list (`references_revised.bib`); `natbib` option `sort&compress` in the preamble.

Response to Reviewer 2

Comment R2 (general)

This paper uses field measurements of conductance and chloride concentrations to demonstrate systematic changes in stream salinity across several mapped high angle faults in New York. The purpose of the study was to test whether the locations of high angle faults correspond to abrupt increases in specific conductance values, which the authors argue would provide evidence that groundwater flow along high angle faults can contribute salt-contaminated groundwater at point locations along streams.

Overall, this study's focus is on an interesting and understudied aspect of streamwater salinity – the authors present preliminary data somewhat consistent with their hypothesis. However, the paper would be strengthened by additional caution with some of their claims; improvements in the clarity of their figures; and additional discussion of alternative interpretations of their data.

Response: We thank the Reviewer for this fair assessment. We agree that the data are preliminary and only partly consistent with the hypothesis, and the revision addresses each of the three areas: the claims are scaled back throughout and the paper is reframed as a hypothesis-generating case study; every figure has been revised as requested; and alternative interpretations are discussed in an expanded Section 4.2 and a new Limitations section (Section 4.4).

Comment R2.1

Introduction. Since many of your contrasts end up being 'confounded' by tributaries, adding a section to the introduction to discuss how tributaries typically influence downstream conductance would set you up to contrast the tributary vs fault boundaries you discuss in your results. Perhaps the tributary dilution effect would be greater if not for the addition of salt-rich groundwater.

Response: We have added a paragraph describing how tributary confluences change main-stem conductance through mixing of two end-members, so that the downstream value depends on the relative discharges and conductivities of the two streams (Genereux, 1998). This sets up the comparison between tributary and fault boundaries in the Results. We also note the Reviewer's suggestion that tributary dilution may be partly offset by salt-rich groundwater, as a possibility that our data cannot test without discharge measurements.

Changes in manuscript:

Section 1: new paragraph on confluence mixing ("We therefore treat tributary and fault boundaries as distinct kinds of contrast"); Section 4.1: dilution offset noted as a possibility that cannot be tested without discharge measurements.

Comment R2.2

Methods. Clarify wheter the normalization was conducted across the data from the whole site or within each zone in each site.

Response: On review, the normalization was not used in any of the statistical tests, which were carried out on specific conductance in $\mu\mathrm{S\,cm^{-1}}$. To avoid confusion we have removed it from the Methods (see also response R1.9), and the figures now show unnormalized values.

Changes in manuscript:

Section 2.3: normalization equation and paragraph removed.

Comment R2.3

On pg 4 you say – 'within-zone scatter is small relative to the between zone contrasts'. Show the standard deviation values to demonstrate this here and throughout the methods and figures (i.e., mean values should have +/- standard error or std, whichever is more statistically appropriate). In addition, somewhere you should show the actual data – if not in a new table 2 in the main text, then at least in an appendix.

Response: Agreed. Zone means are now reported as mean ± standard deviation in the text, in Table 1 and in the zone labels of Figures 2–4. The quoted statement has been removed, because once zones were restricted to single survey days (see R1.7) the comparison it made no longer held in the same form. All individual measurements (field site and point, survey date, distance along channel, position, specific conductance, temperature, zone assignment and field-log flag) are listed in a new Appendix A table and in the public data repository.

Changes in manuscript:

Table 1; Figures 2–4; new Appendix A (all measurements).

Comment R2.4

Results. Pg 5 – It looks like in Fig 3. the fault crosses the stream at several locations, not just where you've mapped in the contrast between two zones. Additional justification/explanation for how the zones (and contrast) were chosen given these additional fault crossings would be helpful.

Response: The Reviewer is right that the fault line in Figure 3 meets the stream in several places. That line is taken from the county-scale map of Budnik et al. (2010; 1:340,000), whose authors describe the fault locations as "largely conjectural since the faults themselves are rarely exposed at the surface" (p. 18). The apparent multiple crossings are therefore not reliable fault locations, and we have removed the crossing markers from Figure 3. The boundary between Zones 2-1 and 2-2 (Contrast B, about 4.0 km along the survey) reflects our field interpretation of where the fault is: an abrupt, roughly 90° bend in the channel, which we took to be the creek turning to follow the fault. Section 2.2 now says so, and notes that the county-scale trace runs along the creek throughout and cannot resolve where the fault crosses the channel. Because an abrupt bend is

consistent with structural control but does not show that groundwater flows along the fault, and because the change-point analysis places the onset of the Site 2 increase about 1.6 km upstream of the bend, Section 4.1 reports Contrast B as an observation worth testing, not as a step at a fault.

Changes in manuscript:

Section 2.2: "At Site 2 the digitized county-scale trace runs along the creek for the whole surveyed reach, so it cannot resolve where the fault crosses the channel. We placed the boundary between Zones 2-1 and 2-2 (Contrast B), about 4 km along the survey, at the position where we interpret the fault to be in the field: an abrupt, roughly 90° turn of the channel, which we took to be the creek turning to follow the fault... An abrupt bend is consistent with structural control of the channel, but it does not by itself show that groundwater flows along the fault there."; Section 3.1: "The digitized county-scale trace runs along the creek throughout and crosses the survey line at about 1.0 and 5.8 km, but those intersections are not reliable fault positions, so we did not use them."; Figure 3 and caption (crossing markers removed; the Contrast B boundary is labelled as the fault position interpreted in the field).

Comment R2.5

The increase in conductance across contrast B is very small – what is the uncertainty in the YSI meter? What was the variability in the readings in each zone? I have seen conductance values change more than 5 us/cm by moving across a stream or from the surface to a deeper point in the water. Without seeing the raw data variability I am skeptical this small difference represents an abrupt change in water source, even if found to be statistically significant after normalization.

Response: We share the Reviewer's caution. Specific conductance was measured with a Thermo Scientific Orion 013010MD conductivity cell, not a YSI instrument; the compatible Orion Star meters specify an accuracy of ±0.5% of reading, about $\pm 1.7\,\mu S\,cm^{-1}$ at Site 2. A calibration offset common to all readings on the same day cancels in a within-day zone difference, whereas random reading error does not. Within-zone standard deviations at Site 2 are 1.1 and 1.9 $\mu S\,cm^{-1}$ (Table 1), and all individual readings are now listed in the Appendix. The tests are on unnormalized values. All readings were taken in the thalweg, so we did not sample the cross-channel or depth variability the Reviewer describes, and we cannot rule out that it contributes to a difference of this size. The change-point analysis also shows that the increase at Site 2 is a gradual rise that begins about 1.6 km upstream of the zone boundary rather than an abrupt change in water source (see R1.5(3)). We therefore describe Contrast B as a small, gradual change near the resolution of the survey, agree that it does not demonstrate a point input at the fault, and list single-point thalweg sampling among the limitations. We also now report that the field log records very high flow at the start of the Site 2 survey, part of which was made by canoe, so that survey may not represent baseflow (Sections 3.1 and 4.4).

Changes in manuscript:

Sections 2.2, 3.1, 3.3, 4.1 and 4.4; Table 1; Appendix A.

Comment R2.6

I am most convinced by contrast E – although by looking at figure 4, the change in conductance value actually occurs within zone 3-3. It's possible that the fault location is slightly off,

correct? If so, could you show that with a fault zone rather than a single line, or at least discuss the possibility that the fault trace is actually east of where it's mapped?

Response: We thank the Reviewer for this careful reading, which, together with Comment R1.7, led us to an error. The jump the Reviewer sees inside Zone 3-3 falls exactly where the 19 October survey ends and the 26 October survey begins: two measurements 18 m apart differ by 23.5 μS cm^{-1} because the stream's overall conductivity changed between the two days, not because of a change along the channel. Compared within a single day, the change across the mapped fault at Contrast E is +1.6 μS cm^{-1} (95% CI −0.3 to +3.4, $p = 0.090$) and not significant. We now report this result and no longer present Contrast E as the strongest evidence (Section 3.1).

On the Reviewer's question about the fault position: within the 26 October survey alone, specific conductance rises by about 4 μS cm^{-1} over the first ∼190 m of the reach and then remains nearly constant (∼360 μS cm^{-1}) for the rest of the reach, including where the channel crosses the digitized fault trace about 520 m along it. The change-point analysis places the breakpoint at 194 m along this reach: 330 m upstream of that crossing, and 123 m in straight-line distance from the nearest mapped trace. We agree that the fault traces carry positional uncertainty, and we looked for a published accuracy with which to draw them as a zone. We did not find one we could cite: the source map is a 1:340,000 county figure that states no positional accuracy, and its authors describe the fault locations as "largely conjectural since the faults themselves are rarely exposed at the surface" (Budnik et al., 2010, p. 18). We therefore draw the traces as lines and discuss the uncertainty in the text rather than show an error zone we cannot support. On the printed scale of the source figure, the breakpoint's 123 m distance from the nearest trace is less than 0.4 mm, so we agree that the trace could lie slightly off its mapped position (as the Reviewer suggests), but our data cannot test this. The rise is also not robust: the breakpoint sits at the minimum segment length the method allows, the rise is not significant as a zone contrast, and it could equally reflect an input unrelated to the fault. The manuscript does not attribute the rise to the fault: it notes that Contrast E is within instrument accuracy and that its only breakpoint lies near the start of a short transect, 330 m upstream of where the channel crosses the trace (Section 4.1), and that we cannot say how far a real fault lies from any mapped position (Section 4.4).

Changes in manuscript:

Sections 3.1, 3.3, 4.1 and 4.4 (Limitations); Figure 4 (survey-date symbology); Table 1. Section 4.4: "so we draw them as lines and cannot say how far a real fault lies from any mapped position."

Comment R2.7

The chloride measurements could be tested for statistically significant difference above and below contrast E – that would be interesting to see as well.

Response: We thank the Reviewer for this suggestion. When we located the survey points flagged for Contrast E on the chloride transect from the GPS positions, we found that the transect ends just downstream of them: only 2 of the 18 main-stem samples lie downstream, too few for a meaningful two-sample test. Chloride near Contrast E is low and flat (the last five samples are 13.75–17.5 mg L^{-1}), so the transect shows no chloride increase there. This also corrects a statement in the original manuscript, which attributed the chloride peak to the Contrast E fault. The three highest values (50, 36.25 and 40 mg L^{-1}) instead lie 133–468 m from the nearest survey point flagged for Contrast D and about 300 m from the nearest digitized fault trace. To describe where chloride

changes along the transect without relying on two-sample tests, we applied the same change-point analysis used for specific conductance. It selects a single breakpoint at 467 m along the transect (bootstrap interval 308–800 m), about 125 m from the nearest Contrast D flagged point and about 300 m from the nearest trace. Because this breakpoint is driven largely by one 50 mg L^{-1} sample, we report it as an observation worth testing, not as support for the hypothesis. Because the chloride survey was made in a different year (July 2020) from the conductivity survey, we present it as a separate observation and do not combine the two.

Changes in manuscript:

Sections 3.2 and 4.1 (chloride results rewritten and reattributed; "The transect therefore shows no chloride increase at Contrast E."); Section 2.2 (transect length 2.35 km); Figure 5 (flagged points for Contrasts D and E and the trace crossing marked along the transect; change-point fit and bootstrap interval).

Comment R2.8

In Zone 3-1 it looks like there are two distinct sets of data overlapping on the figure – a set of higher (370) conductance measurements and a sect of lower (340) conductance measurements. This should be further explained or clarified in the figure caption and main text.

Response: The Reviewer is correct: these are measurements of the same reach on two different days (19 and 26 October 2018) and differ by the day-to-day offset discussed above. Figure 4 now distinguishes the two survey days by symbol, with an inset for the overlapping reach, and both the caption and the text explain the overlap.

Changes in manuscript:

Figure 4 and caption; Sections 2.2 and 3.1 ("Day offset and the repeated reach").

Comment R2.9

Discussion. While some alternative explanations for the increase in conductance at contrasts B and E are discussed in 4.2, landuse differences are dismissed because all sampling was within a conserved forest – however, the maps show distance to roadways and density of roads varies around the various sites and zones within sites. Whether or not the differences in roadway density (and road salting practices) could affect the specific conductance values should be discussed – especially when the observed differences at contrast B are so small. Although you sample in summer, you mention in your introduction that salt can persist in soil for up to 5 months.

Response: Agreed. We have removed the argument that sampling within conserved forest rules out land-use effects. Section 4.2 now discusses how differences in road proximity and road density, and salt stored in roadside soils, could produce conductivity differences of the size observed, especially at Contrast B. To support this discussion we report the median distance from each zone's sampling points to the nearest major road (245–927 m; Table 2); for example, the Site 3 zones lie 245–425 m from the nearest major road, mainly US 44. We also note that our survey was in late September and October 2018 rather than summer, and the text now says so.

Changes in manuscript:

Section 4.2 expanded ("Road proximity": roadside soils and shallow groundwater, and late-season release); Table 2; survey dates stated in Section 2.2 (four days in autumn 2018, Site 1 on 28 September) and Section 4.2.

Comment R2.10

Figures and Tables. Figure 1 (b) is difficult to see details - I would suggest simplifying so just major streams and faults are shown; remove other extraneous details. Or at least highlight the streams in question.

Response: Figure 1b has been simplified to show only major streams, major highways and the mapped high-angle faults on a light basemap, with Wappinger Creek highlighted and labelled, the three study sites boxed, and permitted wastewater facilities marked.

Changes in manuscript:

Figure 1.

Comment R2.11

Figure 2: Label contrast A. Add flow direction.

Response: Done: Contrast A is labelled and a flow arrow is added.

Changes in manuscript:

Figure 2 and caption.

Comment R2.12

Figure 3: Label contrast B. Add flow direction.

Response: Done: Contrast B is labelled and a flow arrow is added. The figure also marks the nearby wastewater facilities; the Contrast B box labels the boundary as the fault position interpreted in the field (see R2.4).

Changes in manuscript:

Figure 3 and caption.

Comment R2.13

Figure 4: Label each contrast E, D, C. Add flow direction. Add margin of error for fault location if known. Add line Z-Z'. Offset or include an inset to better show the overlapping points in Zone 3-2 and 3-1.

Response: Done. Contrasts C, D and E are labelled and flow direction is shown. We searched for a published margin of error for the fault locations but did not find one we could cite: the source map (Budnik et al., 2010) states none and describes the fault locations as "largely conjectural". We therefore show the faults as lines without an error zone, and discuss their positional uncertainty in the text (see R2.6) and in the Limitations section. The chloride transect Z–Z′ is drawn from its GPS positions (yellow band). Survey days are distinguished by symbol (circles and triangles), and an inset enlarges the overlapping reach in Zones 3-1 and 3-2.

Changes in manuscript:

Figure 4 and caption.

Comment R2.14

Figure 5: Either add line Z-Z' in figure 4 or add an inset figure to Fig 5 to show the location of this transect relative to site 3 in the study

Response: Z–Z′ is now shown in Figure 4, and Figure 5 marks the positions of the Contrast D and E flagged survey points, the trace crossing and the Kelly et al. (2019) monitoring site along the transect, together with the change-point fit.

Changes in manuscript:

Figures 4 and 5.

Comment R2.15

Table 1. Show the normalized mean values for each zone. Show some measure of variance for each zone. Consider including chloride measurements above and below contrast E as well

Response: Table 1 now reports each zone's mean ± standard deviation. As explained in our response to R2.7, too few chloride measurements lie downstream of Contrast E for a two-sample comparison, so the chloride results are reported through the change-point analysis and Figure 5 instead. Because the normalization has been removed (see R2.2) and all tests use specific conductance in $\mu\mathrm{S\,cm^{-1}}$, the table reports unnormalized means.

Changes in manuscript:

Table 1; Section 3.2 (chloride).

Response to Reviewer 3

Comment R3.1

Minor Revisions

Response: We thank the Reviewer for their assessment. The revised manuscript addresses the comments of Reviewers 1 and 2, and we hope it also meets this Reviewer's expectations.